

Department of Computer Science

TDT4259 - Applied Data Science

Aneo
Electricity Consumption Forecasting for a Retailer

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1 Problem definition and business understanding

1.1 Context

The Nordic wholesale market, where producers, suppliers and large consumers trade electricity before it reaches end users, consists of three sequential markets: the day-ahead market, the intraday market and the balancing market. The day-ahead market, operated by Nord Pool, is where most of the volume is traded. Between 08:00 and 12:00, participants submit buy and sell bids for each quarter-hour of the following day, and prices are set where supply meets demand, subject to available transmission capacity. After the auction, participants can adjust their positions in the intraday market until one hour before delivery. Remaining deviations are handled by Statnett, which as system operator uses the balancing markets to keep production and consumption in balance in real time. (Energidepartementet 2026)

Electricity retailers buy power in the wholesale market on behalf of their end customers. Since consumption is not known in advance, the retailer submits its day-ahead bids based on a forecast of the next day's consumption. Any deviation between purchased and actual volume is settled as an imbalance at the imbalance price, which may differ from the spot price. (eSett Oy n.d.)

1.2 Problem and opportunity

For an electricity retailer, the quality of the consumption forecast has a direct financial impact. Consumption varies with temperature, calendar effects, time of day and the composition of the customer portfolio, and differs between price areas, while the imbalance price is volatile and hard to predict. Forecast errors can thus lead to substantial and uncertain costs, and improving forecast accuracy represents a direct opportunity for the retailer to operate more cost-effectively in the market.

In this project, we take the role of Aneo's data science team. Aneo delivers consumption forecasts to an electricity retailer under a fixed-price contract. The task is to forecast the retailer's hourly consumption for each hour of the next day in six price areas (NO1-NO5 and FI). The forecast must be delivered around 10:00, so that the retailer can use it in its day-ahead bids before the auction closes at 12:00. From a data science perspective, this is a regression problem consisting of predicting hourly consumption for the next day, using only information available at 10:00 on the day of prediction. As a baseline, we assume consumption to be equal to the same hour one week earlier.

1.3 Stakeholders and decision

Three stakeholders are considered: Aneo as the forecast provider, the retailer as the end user of the forecast, and indirectly the retailer's customers. The retailer makes the final decision on how much electricity to buy in the day-ahead market, and the forecast serves as a decision-support tool rather than an automated decision. Because Aneo operates under a fixed-price contract, they do not bear the imbalance costs directly. The value of an accurate forecast for Aneo is therefore indirect: lower imbalance costs for the retailer increase customer satisfaction and the likelihood of contract renewal. Aneo, in turn, must decide which forecaster to deliver in each price area and whether the service is worth offering at the agreed fee. The key business question is therefore not only which model predicts consumption most accurately, but whether the solution is reliable and valuable enough for the retailer to keep using it.

1.4 Success criteria

Before developing a model, success criteria must be defined in order to assess its value. The criteria are divided into four tiers: forecast quality, business value, operational reliability and customer relationship.

- **Tier 1: Forecast quality**
  These metrics measure how much the forecast deviates from actual consumption. All metrics are evaluated on a chronological test period.
  Let y_t denote actual consumption and $\hat{y}_t$ the forecast in hour t, with n hours in the evaluation period:

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t| \quad (1) \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2} \quad (2)$$

$$\text{WAPE} = \frac{\sum_t |y_t - \hat{y}_t|}{\sum_t y_t} \quad (3) \qquad \text{NMBE} = \frac{\sum_t (\hat{y}_t - y_t)}{\sum_t y_t} \quad (4)$$

Table 1: Forecast accuracy metrics (Tier 1).

Metric	What it measures	Relevance in this case
MAE	Average absolute error (MWh)	Primary metric. With symmetric loss, it directly represents the average imbalance per hour.
RMSE	Error with squared deviations (MWh)	Penalises large errors more heavily, revealing occasional severe misses.
WAPE	Absolute error relative to total consumption (%)	Makes performance comparable across price areas with different consumption levels.
NMBE	Signed error relative to total consumption (%)	Reveals systematic over- (> 0) or underforecasting (< 0). Target: close to zero.

All metrics are also computed for the baseline (same hour one week earlier), and the model is considered successful only if it outperforms the baseline both overall and in each price area, in which the model is applied. In addition to aggregate values, MAE is reported by hour of day, by price area and for high consumption periods, to reveal whether errors are concentrated when imbalances are most costly.

Forecast accuracy is, however, only a proxy for business value, which is addressed in Tier 2.

- **Tier 2: Business and financial value**

  Although Aneo operates under a fixed-price contract, reducing the retailer's imbalance costs increases customer satisfaction and the likelihood of contract renewal (see section 1.3). Business value is therefore assessed with the following KPIs:

  Notation: For price area a (NO1–NO5 and FI) and hour t, $y_{a,t}$ is actual consumption (MWh), $\hat{y}^{\text{model}}_{a,t}$ the model forecast, $\hat{y}^{\text{base}}_{a,t} = y_{a,t-168}$ the baseline (same hour one week earlier) and $p^{\text{spot}}_{a,t}$ the spot price (NOK/MWh). Sums run over all areas and hours in the evaluation period.

  – *Total imbalance volume:* total physical imbalance volume the retailer must settle on the balancing market over the evaluation period.

$$\text{Total Imbalance} = \sum_a \sum_t |\hat{y}^{\text{model}}_{a,t} - y_{a,t}| \quad (5)$$

  – *Avoided imbalance volume*: physical volume mismatch prevented by using the ML model instead of the baseline.

$$\text{Avoided Imbalance Volume} = \sum_a \sum_t |\hat{y}^{\text{base}}_{a,t} - y_{a,t}| - \sum_a \sum_t |\hat{y}^{\text{model}}_{a,t} - y_{a,t}| \quad (6)$$

  – *Avoided imbalance cost (savings):* the value (NOK) of the reduction in absolute forecast error compared with the baseline. For each price area a and hour t, the reduction in error is weighted by the spot price, so errors in high-price hours count more:

$$\Delta C = \sum_a \sum_t \left(|\hat{y}^{\text{base}}_{a,t} - y_{a,t}| - |\hat{y}^{\text{model}}_{a,t} - y_{a,t}| \right) \cdot p^{\text{spot}}_{a,t}, \quad (7)$$

  A positive ΔC means the model has a lower price-weighted error than the baseline. Since imbalance prices are not available (see Section 1.5), the spot price is used as a proxy for the cost per MWh of a forecast error. The metric assumes that over- and underforecasting cost the same per MWh.

  – *P95 imbalance:* the absolute error that is exceeded in only 5% of hours. While MAE describes the typical error, P95 describes the size of the errors in the worst hours, indicating how much imbalance the retailer should be prepared to handle. Assuming unbiased, normally distributed errors, P95 $\approx$ 2.5·MAE where a higher ratio indicates more frequent extreme errors than this assumption implies. A low P95 therefore indicates lower risk of large, unexpected imbalance costs, which is relevant for how the retailer prices its customer contracts (see Tier 4). Target: a lower P95 than the baseline in every area where a model is deployed.

$$\text{P95 Imbalance} = \text{Percentile}_{95}\left(\left\{ |\hat{y}^{\text{model}}_{a,t} - y_{a,t}| \right\}_{a,t} \right) \quad (8)$$

– *Net Service ROI*: quantifies the return on investment for retailer. It compares saved imbalance costs against Aneo's service fee. The contract fee is not given in the case. Instead, the simulated savings are used to indicate the maximum fee the retailer would be willing to pay for the service. For the service to be viable, this must exceed both a reasonable market price for forecasting services and Aneo's own cost of developing and operating the model. We assume a target of ROI $\geq 200\%$, i.e. savings of at least three times the cost. The margin is set high to account for uncertainty in the savings estimate, which relies on spot prices as a proxy for imbalance costs.

$$\text{ROI} = \left(\frac{\text{Avoided Cost} - C_{\text{contract}}}{C_{\text{contract}}} \right) \times 100 \tag{9}$$

- **Tier 3: Operational reliability**
  The pipeline must use no consumption data newer than 144 hours (the 5-day metering delay plus a safety margin, section 2.5), so no future information leaks into training or inference. Also, the forecast must be delivered by 10:00 local time every day, including weekends and holidays, with a target of 100% on-time delivery. If the model fails or input data is missing, the baseline forecast is delivered automatically, since a simple forecast is better than none. Monitoring and retraining are described in Section 6.
- **Tier 4: Customer relationship**
  In the longer term, the success of the service is reflected in retailer satisfaction and contract renewal. These cannot be calculated directly from the dataset, but follow from the KPIs above:
  – *Value:* a more accurate forecast leads to smaller imbalances and lower financial exposure, increasing the perceived value of the service.
  – *Risk:* large imbalances in a few hours can cause substantial and unpredictable costs. For fixed-price contracts, this risk is covered by a risk premium in the end-customer price. A lower P95 reduces this exposure, allowing the retailer to offer more competitive and predictable prices.
  – *Interpretability:* since the retailer combines the forecast with its own market assessments, the main drivers, such as temperature and calendar effects, should be visible.
  – *Trust:* share of hours where the relative forecast error is within $\tau = 15\%$. We assume this is the largest error a trader would accept in a single hour without questioning the forecast. Target: Trust $\geq 90\%$ overall and higher than the baseline.

$$\text{Trust} = \frac{100}{N} \sum_{a} \sum_{t} \mathbb{1}\left(\frac{|\hat{y}_{a,t}^{\text{model}} - y_{a,t}|}{y_{a,t}} \leq \tau \right) \tag{10}$$

  where N is the number of area-hours and $\mathbb{1}(\cdot)$ equals 1 if the condition holds and 0 otherwise. Hours with zero consumption are excluded. Together with reliable delivery (Tier 3) and stable performance across price areas, this builds the retailer's trust in, and continued use of, the forecasts.

1.5 Constraints and scope

The success criteria above define what the forecasting solution should achieve. Achieving them is, however, subject to several practical, temporal and business constraints, which also define the scope of the project.

– **Forecast horizon and information:** The forecast is made at 10:00 for 14-37 hours ahead, using only information available at that time. Temperature inputs must therefore be weather forecasts, not observed values.
– **Delayed consumption data:** Metered consumption is available with a delay of about five days, so the most recent consumption lies at least six days before the target day.
– **Data scope:** The data covers one retailer's customer portfolio, not total consumption in each price area. Consumption may shift as the retailer gains or loses customers, and results may not generalise to other retailers.
– **Simplifications:** Forecasts are made at hourly resolution, although the day-ahead market has used 15-minute resolution since 2025 (Energidepartementet 2026). Historical spot prices are available, but imbalance prices are not. The imbalance cost in Equation (7) is therefore approximated.
– **Out of scope:** The project forecasts consumption only. Price forecasting is excluded, as prices are set by the market and are not part of Aneo's service. The retailer's bidding strategy is excluded, since the retailer makes the final purchasing decision, and so is Aneo's own production planning.

2 Data understanding and background

2.1 Domain background and structural dynamics

Electricity consumption in the Nordic wholesale power market is driven by physical climate conditions, national heating infrastructures, and macroeconomic shocks (Energidepartementet 2026). Understanding these mechanisms is crucial to design effective features and avoid model misspecification across price areas.

Space heating infrastructure disparity. In Norway, electricity demand is exceptionally weather-sensitive. Driven by historically abundant hydropower, direct electric resistance heaters and heat pumps account for the vast majority of residential and commercial space heating (Statistisk sentralbyrå 2023a). Consequently, winter sub-zero temperatures trigger dramatic load surges across Norwegian bidding zones (NO1-NO5). Conversely, urban Finland (FI / Helsingfors) relies predominantly on municipal district heating networks (*kaukolämpö*), which supply over 90% of apartment buildings with thermal energy from combined heat and power plants (Helen Ltd 2024). As a result, electricity consumption in Helsingfors is largely decoupled from ambient temperature, exhibiting a much flatter annual profile and minimal cold-weather elasticity.

Macroeconomic context and activity cycles. The dataset period (April 2022 to April 2023) coincided with the acute phase of the European energy crisis following the Russian invasion of Ukraine and subsequent gas supply disruptions (International Energy Agency 2022). Elevated continental electricity and fuel prices spilled into interconnected southern Norwegian bidding zones (NO1, NO2, NO5), inducing voluntary demand conservation and behavioral load adjustments. Simultaneously, regular human routines produce deterministic intra-day diurnal curves: a morning ramp (06:00-09:00 UTC), an evening peak (17:00-20:00 UTC), and a night trough (01:00-04:00 UTC). While activity drops slightly on off-days (weekends and public holidays), empirical analysis reveals that seasonal temperature swings vastly dominate calendar-day variance.

2.2 Data sources and descriptive statistics

The primary dataset provided for this project comprises hourly records from April 2022 to April 2023 across six Nordic bidding zones: NO1 (Oslo), NO2 (Stavanger), NO3 (Trondheim), NO4 (Tromsø), and NO5 (Bergen) in Norway, plus FI (Helsingfors) in Finland. It contains exclusively two variables: **metered consumption (y_t, MW)** and **temperature forecasts (T_t, °C)**. y_t represents the realized hourly electricity consumption for the specific retailer's customer portfolio, while T_t is the hourly temperature forecast assigned to each bidding area.

To evaluate the business and financial impact defined in Tier 2 (simulating avoided costs in Equation 7), we independently augmented the dataset by collecting historical hourly day-ahead spot prices (p_t^{spot}) from the ENTSO-E Transparency Platform (ENTSO-E 2023), converted to NOK using official exchange rates (Norges Bank 2023). While using an average price proxy was an alternative, actual hourly spot prices significantly enhance financial fidelity without introducing modeling overhead.

Table 2 reports descriptive statistics across bidding areas. Portfolio volume varies substantially: Oslo accounts for the highest load (mean 6.88 MW, peaking at 18.05 MW), whereas Tromsø is the smallest (mean 1.34 MW). Stavanger's demand exceeds Trondheim's, reflecting the retailer's regional market share rather than city population (Trondheim $\approx 212,000$ vs. Stavanger $\approx 145,000$ inhabitants in 2022 and 2023, (Statistisk sentralbyrå 2023b)).

Table 2: Descriptive statistics of hourly consumption (MW) and temperature (°C) across bidding zones.

			Consumption (MW)			**Temperature (°C)**		
Zone / City	**Period**	**Obs.**	**Mean ± Std**	**Min**	**Max**	**Mean ± Std**	**Min**	**Max**
NO1 / Oslo	07/04/22–02/04/23	8,641	6.88 ± 4.67	1.52	18.05	7.39 ± 8.77	-15.4	29.8
NO2 / Stavanger	07/04/22–02/04/23	8,641	3.30 ± 2.11	0.83	8.76	8.55 ± 5.84	-9.2	27.0
NO3 / Trondheim	07/04/22–02/04/23	8,641	2.28 ± 1.14	0.62	5.10	6.21 ± 7.62	-17.3	30.5
NO4 / Tromsø	07/04/22–02/04/23	8,641	1.34 ± 0.56	0.37	2.55	3.51 ± 7.97	-14.6	29.8
NO5 / Bergen	07/04/22–02/04/23	8,641	1.88 ± 1.16	0.44	4.65	7.75 ± 6.56	-15.6	29.3
FI / Helsingfors	14/07/22–02/04/23	6,289	6.08 ± 0.92	0.00	7.86	5.66 ± 7.59	-10.1	24.0

2.3 Exploratory data analysis and important variables

Scale disparities and seasonal trajectory. Figure 1 highlights large scale differences across price areas, with Oslo having by far the highest volume and Tromsø the lowest. All Norwegian series display a clear U-shaped seasonal pattern, where winter demand doubles or triples compared to summer due to electric heating, alongside regular 24-hour daily cycles. In contrast, Helsingfors immediately stands out as an anomaly: its telemetry starts three months later and its load remains remarkably flat year-round, showing no winter heating surge at all.

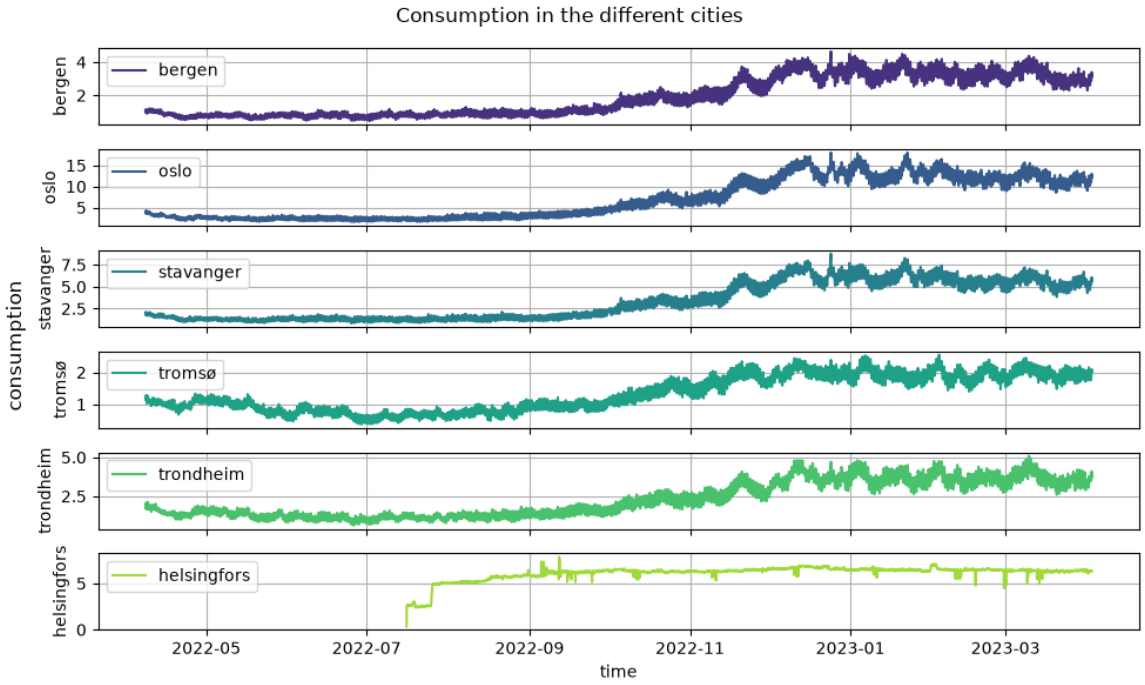


Figure 1: Hourly electricity consumption series across bidding zones (April 2022 – April 2023).

Non-linear heating curves and Finnish divergence. Figure 2 presents the city-by-city scatter plots of consumption against temperature. This reveals a clear non-linear heating curve: above 15 °C, electricity demand remains flat at baseload levels, confirming negligible air conditioning in the Nordics. Below 15 °C, demand rises linearly, accelerating sharply below 0 °C as electric heating operates near peak capacity. Crucially, the city-by-city breakdown illustrates that Helsingfors is completely insensitive to sub-zero temperatures (narrow IQR = 0.28 MW), empirically validating the dampening role of municipal district heating.

Important variables and feature drivers. Guided by exploratory findings and operational constraints, the information used for modeling is grouped into key conceptual drivers (whose detailed feature construction is described in Section 3.2.2):

- *Target and location:* hourly metered consumption (y_t, MW) and bidding area identifiers.
- *Calendar and temporal signals:* hour of day (capturing morning and evening demand peaks), day of week, month, and indicator flags for weekends and national holidays.
- *Historical load signals:* autoregressive consumption lags respecting the 5-day latency (144h, 168h, 192h, 336h) and 7-day rolling statistics to establish baseline portfolio scale and load variability.
- *Weather forecasts and thermal dynamics:* forecasted temperature, daily mean temperatures, 24-hour temperature changes, and thermal differences aligned with weekly load lags.
- *Non-linear weather terms:* while gradient boosting tree models (LightGBM) naturally segment non-linear thresholds, linear models like Ridge Regression require explicit terms (namely: squared temperature T^2 and heating degree days below 15 °C) to capture the physical heating inflection.

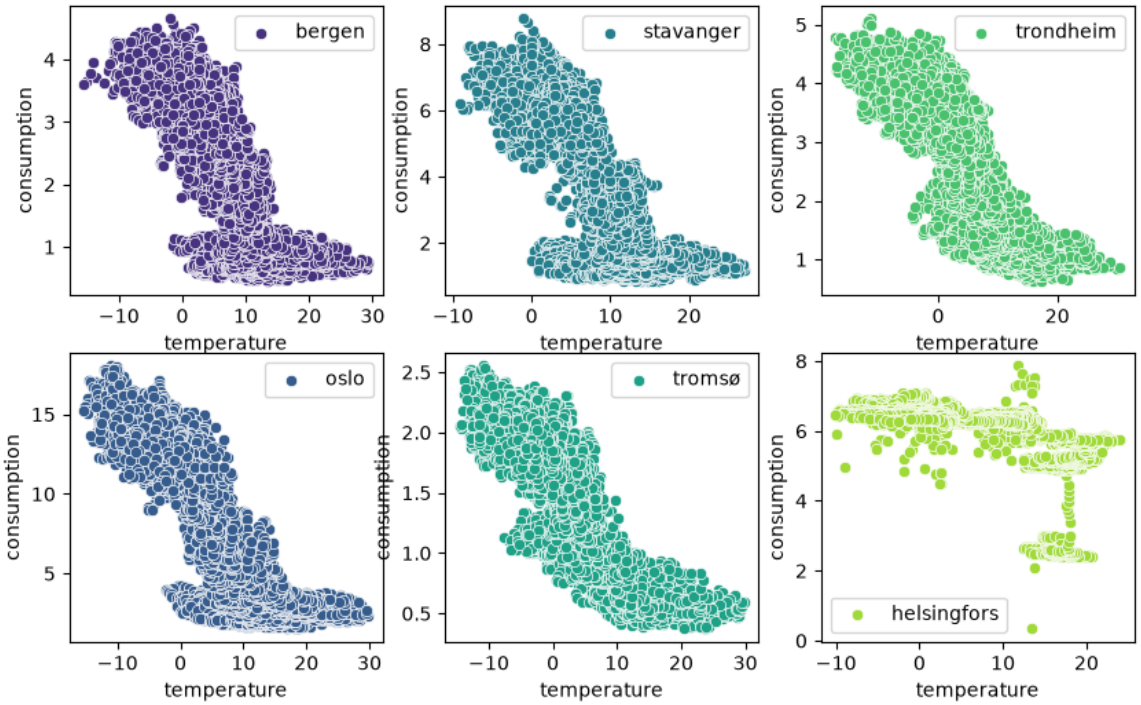


Figure 2: Consumption versus temperature by bidding zone

2.4 Data quality issues, anomalies and remediation

Rigorous screening was performed to verify data integrity and isolate actionable quality issues:

- **Limited temporal sample:** The dataset spans only 12 months, observing only a single winter cycle and precluding multi-year seasonal cross-validation.
- **Asymmetric temporal coverage.** Norwegian zones contain 8,641 consecutive hours (April 7, 2022 to April 2, 2023), whereas Helsingfors contains only 6,289 records, commencing three months later on July 14, 2022, probably due to subsequent customer onboarding.
- **Absence of internal missing records.** Once telemetry was active, zero missing timestamps or NaN values occur across all six locations in any hour.
- **Physical extremes vs. spurious outliers.** Statistical screening ($1.5 \cdot \text{IQR}$ and 3σ) flagged Oslo's winter peaks ($\sim$ 18 MW, on Christmas Eve and -12 °C freeze) as global outliers. However, domain validation confirms these represent authentic space heating demand. Removing or winsorizing them would dangerously understate balancing risks in Tier 2 financial metrics. Likewise, thermal extremes (-17.3 °C in Trondheim, +30.5 °C in summer) are verified records essential for training weather elasticity.
- **Telemetry initialization artifact in Helsingfors.** Exactly 32 consecutive observations report consumption = 0.0 MW in Helsingfors between 2022-07-14 21:00 and 2022-07-16 04:00 UTC at the very beginning of the series, reflecting telemetry initialization latency.
  Remediation. Rather than imputing synthetic values, we trim the Helsingfors start date to 2022-07-16 05:00:00 UTC, ensuring complete, unbroken hourly records across all areas.
- **Missing weather forecast metadata.** Temperature is provided as a forecast, but without metadata indicating when it was issued. In real-world operations, older forecast runs carry higher errors, and the pipeline must guarantee that the latest weather update is delivered in time for the 10:00 nomination deadline.

2.5 Assumptions and operational limitations

The forecasting task operates under several binding assumptions and practical limitations:

- **Operational 5-day metered latency.** At 10:00 on day D, the retailer bids for all 24 hours of day $D+1$ (lead

times 14 to 37 hours ahead). The case specifies an 'approximate 5-day delay' for verified metered consumption. Taking an exact 5-day (120h) cutoff from the 10:00 decision deadline would theoretically allow a 134-hour lag for the earliest target hour (120h + 14h = 134h at 00:00 on $D + 1$). However, because meter readings in power utilities are processed in daily settlement batches and delivery schedules fluctuate, relying on 134 hours carries a severe risk of operational data leakage if verification runs even slightly late. Following discussions on this design choice, we adopted a conservative, full 144-hour lag ($\tau \geq 144$ h, exactly 6 days prior at the same hour). This guarantees that for all 24 target hours, consumption observations are fully settled and observable at 10:00 under real-world conditions. This constraint eliminates standard short-term autoregression (lag_{24}, lag_{48}) and immediate error feedback, forcing models to rely primarily on weekly seasonality (lag_{168}) and weather forecasts.

- **Single-year horizon and seasonal generalizability.** Spanning only one annual cycle, the dataset captures a single winter. Models cannot assess performance across varying winter severities (mild vs. severe winters) or decouple 2022 energy crisis shocks from permanent structural trends.
- **Unforeseeable external and extreme events.** Purely empirical machine learning models cannot anticipate unforeseen geopolitical interventions (such as regulatory price caps or export restrictions), major transmission grid outages, or unprecedented extreme weather shocks not represented in historical training data.
- **Unknown customer portfolio changes.** The data shows only total electricity consumption, without providing the number of customers or their type (households vs. businesses). If the retailer gains or loses clients over time, the baseline electricity demand will shift. Because models cannot observe customer counts, they may mistakenly confuse these business portfolio changes with model forecasting errors.
- **Meteorological parsimony.** Temperature is the sole weather variable provided. Other thermodynamic drivers affecting building heat loss and heat pump efficiency (for instance, wind speed, solar irradiance and relative humidity) are unobserved.
- **Deterministic weather forecast assumption.** Temperature forecasts are assumed known and error-free at nomination time, neglecting weather forecast error propagation over the 14-37 hour lead horizon.
- **Market price-taker assumption.** In simulated imbalance cost evaluations, the retailer's forecasting errors are assumed not to influence day-ahead or balancing market clearing prices.

3 Methods, analysis and technical evaluation

3.1 Overview and methodological framework

Problem formulation. As defined in Section 1.2, the task is a supervised regression problem on a panel of six hourly time series. For each price area a and each hour t of the target day $D+1$, the model estimates

$$\hat{y}_{a,t} = f\left(\mathbf{x}_{a,t}\right), \qquad \mathbf{x}_{a,t} = \left[\, \mathbf{c}_t,\ \mathbf{w}_{a,t},\ \mathbf{l}_{a,t-\tau},\ a \,\right], \quad \tau \geq 144 \text{ h}, \tag{11}$$

where $\mathbf{c}_t$ are calendar features, $\mathbf{w}_{a,t}$ the temperature forecast and its derivatives, and $\mathbf{l}_{a,t-\tau}$ lagged consumption statistics. Only information available at the nomination time (10:00 on day D) enters $\mathbf{x}_{a,t}$. The last metered value available at that time is the one for 23:00 on day $D-5$ (Section 2.5), which lies 121 hours before the first and 144 hours before the last target hour. Requiring $\tau \geq 144$ therefore makes every consumption-based feature observable for all 24 target hours, so the offline setup is identical to the operational one. Temperature forecasts for day $D+1$ are available at nomination time and enter without lag. Unlike classical short-term load forecasting, where the most recent load is the strongest predictor (Hong and Fan 2016), the most recent load used here is already six days old, so the forecast relies on weather, calendar and consumption patterns from earlier weeks, and all 24 hours are forecast in a single direct step (Ben Taieb et al. 2012).

Design principles. Three principles guide the methodological choices in this chapter.

- *No information leakage.* Every feature, rolling statistic and validation fold must be computable at 10:00 on day D. Violations inflate offline accuracy and are a well-known source of failure at deployment (Kaufman et al. 2012).
- *Parsimony relative to data size.* The dataset spans one year only (Section 2.4), so we favour models whose capacity can be controlled and whose behaviour can be inspected over deep architectures, which typically require much more data. Recent evidence shows that simple models can match or outperform complex ones in long-horizon forecasting (Zeng et al. 2023), and that tree ensembles remain highly competitive on tabular data of moderate size (Grinsztajn et al. 2022).
- *Evaluation against a transparent baseline.* Every model is compared with a seasonal-naive forecast, using the metrics of Table 1, in line with standard forecasting practice (Hyndman and Athanasopoulos 2021).

Forecasters compared. We compare a seasonal-naive baseline ($\hat{y}_{a,t} = y_{a,t-168}$, also the operational fallback of Tier 3 in Section 1.4), a global LightGBM model (Ke et al. 2017) as main model, and local Ridge regressions (Hoerl and Kennard 1970) as a simpler linear alternative. They are detailed in Sections 3.3 and 3.4.

Global versus local modelling and treatment of scale. A global model fitted on the whole panel shares information across series and is often competitive with, or better than, one model per series, particularly when each series is short (Montero-Manso and Hyndman 2021), local models are specific to each series. The areas differ widely in scale (mean consumption from 1.34 MW in NO4 to 6.88 MW in NO1, Table 2). Decision trees are invariant to monotone transformations of the inputs (Hastie et al. 2009) and the global LightGBM separates the level of each area through the area feature, so no normalisation is applied and the target stays in MW, meaning each local Ridge model works on its own area's scale. Because the squared-error loss weights areas in proportion to their scale, we report the scale-free WAPE per area in addition to the aggregate MAE (Section 3.6). The two strategies differ in both pooling and model class, a limitation discussed in Section 3.7.

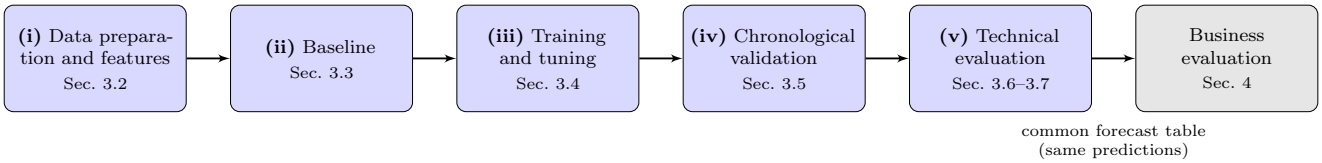


Figure 3: Methodological pipeline of the chapter.

3.2 Data preparation and feature engineering

3.2.1 Data preparation

The modelling table is in long format, with one row per price area and hour and the columns `time` (UTC), `location`, `consumption` (MW) and `temperature` (°C). The quality checks and the only remediation applied, trimming the Helsingfors series to start on 2022-07-16 05:00 UTC to remove the zero-valued initialisation records, are described in Section 2.4 where no imputation is needed. Extreme winter peaks are retained, as they are genuine heating demand. The Norwegian areas and Helsingfors are kept as separate areas, since their temperature sensitivity differs structurally (Section 2.1).

Leakage controls. The first design principle is implemented as follows:

- Consumption-based features (lags and rolling statistics) are computed *within* each price area and shifted by at least 144 hours, so that no metered value after 23:00 on day $D-5$ enters them.
- Temperature features may refer to the target hour itself, since temperature forecasts for day $D+1$ are available at nomination time (assumed error-free, Section 2.5).
- Rows with missing lags at the beginning of each series are dropped, without imputation.
- The standardisation of the Ridge inputs is fitted on the training period only.

The chronological split and the tuning procedure are described in Sections 3.5 and 3.4.

3.2.2 Feature engineering

The feature set follows from the exploratory analysis (Section 2.3) and from the information constraint of Section 3.1. Table 3 lists all features and states why each is available at 10:00 on day D. The two models share the same base features; the heating-degree and squared-temperature terms are added for the Ridge models only, for the reason given below.

Table 3: Feature set. t is the target hour on day $D+1$, T the temperature forecast and y metered consumption.

Group	Feature	Definition	Available
Calendar	`hour`, `dow`, `month`	Hour of day, day of week and month of t	Known
	`is_weekend`, `is_holiday`	Weekend flag; public holiday in the area's country	Known
Weather	`temperature`	T_t	Forecast
	`temp_day_mean`	Mean temperature forecast of the day of t	Forecast
	`temp_lag_24`	T_{t-24}	Forecast
	`temp_diff_144`, `temp_diff_168`	$T_t - T_{t-144}$, $T_t - T_{t-168}$	Forecast
Autoregressive	`lag_144`, `lag_168`, `lag_192`, `lag_336`	y_{t-k} for $k \in \{144, 168, 192, 336\}$	Metered (≥ 144 h)
	`roll_mean_7d`, `roll_std_7d`	Mean and standard deviation of y over the 168 hours ending at $t-144$	Metered (≥ 144 h)
Area	`location`	Price area, categorical (global LightGBM only)	Known
Ridge only	HDD_t, T_t^2	$\max(15 - T_t, 0)$ and T_t^2	Forecast

Calendar features. Electricity demand follows deterministic daily and weekly routines, with a morning ramp, an evening peak and a night trough (Section 2.1). Hour of day and day of week capture these cycles, `is_weekend` and `is_holiday` mark off-days, using the public-holiday calendar of the country of each area, and the month captures slower seasonal variation not fully explained by temperature, such as daylight hours and building usage. Calendar effects are a core component of load models (Hong and Fan 2016). Although weekday and off-day load distributions overlap to a large extent and temperature dominates day-type variation (Section 2.3), the flags are inexpensive and let the models capture the residual effect of off-days.

Temperature features. Besides the forecast T_t, we include the daily mean forecast, which reflects the heating

need over the whole day, the temperature 24 hours earlier, and the differences between the target-hour temperature and the temperature 144 and 168 hours earlier. The latter two are paired with the consumption lags of the same length and let the model correct the lagged load for the change in weather. Temperature is the only weather variable available (Section 2.5).

Since consumption rises non-linearly below about 15 °C (Figure 2), the linear Ridge models additionally receive T_t^2 and the heating-degree term $\mathrm{HDD}_t = \max(15 - T_t, 0)$ (Sailor and Muñoz 1997), with the threshold taken from the exploratory analysis. LightGBM does not need them: both are functions of T_t alone, which a tree ensemble reproduces by splitting on T_t, so they would be redundant (Hastie et al. 2009).

Autoregressive features. The data delay removes the short lags that dominate most day-ahead load models (lag 24 and lag 48), so the autoregressive part relies on information at least 144 hours old. The 168-hour lag carries the daily and weekly pattern, the 144- and 192-hour lags describe the neighbouring days, and the 336-hour lag adds a second weekly observation that makes the profile robust against one-off deviations. The rolling mean and standard deviation of consumption over the 168 hours ending 144 hours before the target hour,

$$\bar{y}_t^{7d} = \frac{1}{168} \sum_{k=144}^{311} y_{t-k}, \qquad s_t^{7d} = \sqrt{\frac{1}{167} \sum_{k=144}^{311} \left(y_{t-k} - \bar{y}_t^{7d}\right)^2}, \tag{12}$$

summarise the recent level and volatility of the portfolio, which can drift as customers join or leave. All these features are computed within each price area, with the shift applied before the rolling window.

The month feature. The dataset covers a single year, so each calendar month is observed once and the test period (January to early April 2023) falls largely in months absent from the training period (April to December 2022). The feature can therefore only act as a proxy for the specific winter observed and for the 2022 energy crisis (International Energy Agency 2022), not as a learned seasonal pattern. We tested the models with and without it: removing it worsens the LightGBM forecasts, while the Ridge model improves. We keep the original feature set for both models and report the sensitivity of the Ridge model as a technical limitation of the single-year dataset (Section 3.7) with several years of data, a smooth annual seasonality would be preferable.

3.3 Baseline: seasonal naive forecast

The baseline repeats the consumption of the same hour one week earlier,

$$\hat{y}_{a,t}^{\text{base}} = y_{a,t-168}, \tag{13}$$

the standard benchmark for series with a strong seasonal pattern (Hyndman and Athanasopoulos 2021). Since $168 \geq 144$, it respects the data-availability constraint, needs no training and is applied to each area separately. It is a useful reference because it is transparent and costs nothing to run: any model must beat it, overall and in each price area where it is applied (Section 1.4), to justify its complexity, and it is the fallback delivered if the main model fails (Tier 3, Section 1.4). It is also a poor predictor by construction: it uses no weather information, and it reproduces any anomaly of the reference week, such as a public holiday or a cold spell, while missing the change in weather between the two weeks.

3.4 Model training and hyperparameter tuning

Both models use the same base features (Section 3.2.2) and are trained on the data up to 144 hours before the start of the 90-day test period (Section 3.5) and all tuning takes place inside this training period. The models are implemented in Python with `scikit-learn` (Pedregosa et al. 2011), `LightGBM` and `Optuna`, and the random seed is fixed (42). The settings of both models are given below. The test forecasts of both models are stored in a common table, which is the input to the business evaluation (Section 4).

Cross-validation scheme. Hyperparameters are selected with an expanding-window time-series cross-validation with four folds (Bergmeir and Benítez 2012) where each fold is fitted on all data up to a cut-off and validated on the following block, so validation always lies after the training data. A gap of at least 144 hours separates the two, mirroring the metering delay. Both models use this scheme, with the MAE as selection criterion (Table 1).

LightGBM. The global model is a gradient-boosted tree ensemble (Friedman 2001; Ke et al. 2017) trained with squared-error loss on all six areas. Nine settings are tuned with Optuna (Akiba et al. 2019), using the

Tree-structured Parzen Estimator sampler (Bergstra et al. 2011) over 40 trials. In each trial the model is fitted on the four folds with early stopping on the validation MAE, the trial score is the mean validation MAE and the number of trees is the median of the best iterations. Table 4 gives the search space and the selected values (cross-validated MAE 0.551 MW). The ranges combine row and column subsampling with ℓ_1 and ℓ_2 penalties to control overfitting on one year of data. The selected depth limit of four allows at most 16 leaves, so depth, not the maximum number of leaves, constrains complexity. The final model is refitted on the whole training period.

Table 4: LightGBM hyperparameters: search space and selected values.

Hyperparameter	**Search range**	**Selected**
Learning rate	[0.01, 0.1] (log scale)	0.071
Number of leaves	$\{15, \ldots, 127\}$	80
Maximum depth	$\{4, \ldots, 12\}$	4
Minimum samples per leaf	$\{10, \ldots, 100\}$	75
Row subsampling	[0.6, 1.0]	0.84
Column subsampling	[0.5, 1.0]	0.55
ℓ_1 penalty (`reg_alpha`)	[10^{-3}, 10] (log scale)	0.014
ℓ_2 penalty (`reg_lambda`)	[10^{-3}, 10] (log scale)	0.031
Number of trees	early stopping (maximum 2000)	502

Ridge regression. The alternative is one ridge regression (Hoerl and Kennard 1970) per price area, with the same features plus the squared temperature and heating-degree terms (Section 3.2.2), and without the area variable. Hour, weekday and month are one-hot encoded and the remaining inputs are standardised, since the ridge penalty depends on the scale of the inputs (Hastie et al. 2009). The only hyperparameter, the penalty strength α, is selected for each area among 25 logarithmically spaced values between 10^{-3} and 10^{3} with the scheme above, and the model is refitted on the whole training period of that area. The selected values range from 31.6 (Oslo) to 316 (Helsingfors).

3.5 Chronological validation and test design

Hold-out test period. The last 90 days of the sample are held out as test period, from 2 January 2023 22:00 to 2 April 2023 21:00 UTC (2,160 hourly observations in each of the six areas, 12,960 in total). The training set ends 144 hours earlier, on 27 December 2022 21:00 UTC, and contains 33,622 rows. Because the features need up to 14 days of history, it effectively starts on 21 April 2022 for the Norwegian areas (6,001 hours each) and on 30 July 2022 for Helsingfors (3,617 hours). The test period contains only the cold season, which limits what the results say about the rest of the year (Section 3.7).

Evaluation protocol. In the main evaluation each model is fitted once on the training set and used for all 90 days, which gives every forecaster the same training data and requires tuning only once. The features of every test hour are computed from observed consumption at least 144 hours old, as in operation, so only the model parameters are fixed. Since the retraining schedule is a design choice (Section 5.4), the effect of updating the models is tested in a separate experiment (Section 3.6). The baseline needs no fitting.

Hyperparameters are tuned only within the training period, with the expanding-window scheme of Section 3.4 (Figure 4).

3.6 Technical evaluation

This section reports the forecast-quality metrics of Table 1 on the test period. The business metrics derived from the same forecasts are reported in Section 4.

Overall accuracy. Table 5 summarises the results over all areas. Both models clearly improve on the baseline, which has an MAE of 0.525 MW and a WAPE of 9.5%. LightGBM reduces the MAE by 45% (to 0.288 MW, WAPE 5.2%) and Ridge by 42% (to 0.305 MW, WAPE 5.5%). LightGBM is also better than Ridge on the other metrics: its MAE is 5.6% lower, its RMSE 13% lower (0.426 MW against 0.489 MW), which means fewer large errors, and its bias is close to zero (NMBE +0.3%), whereas Ridge underforecasts on average (−2.6%). Since a single LightGBM fit depends on the random seed (Section 3.7), this difference in a single run should not be over-interpreted.

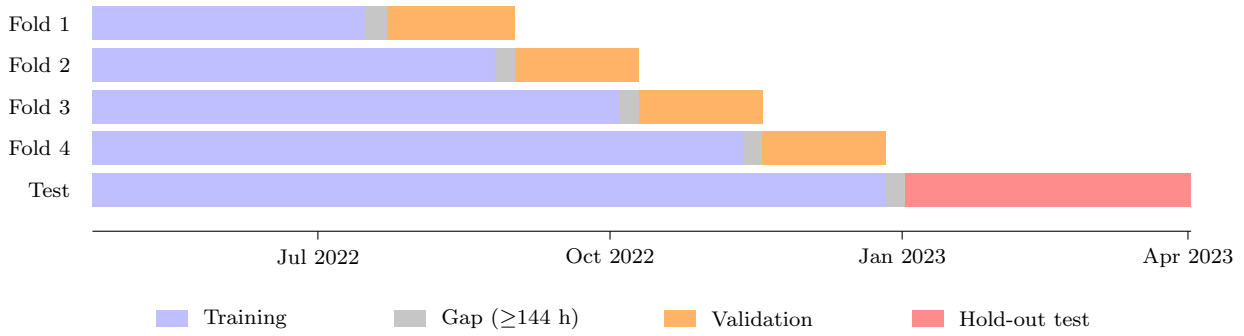


Figure 4: Training, validation and test periods (pooled panel). The last row shows the final fit on the whole training period and the 90-day hold-out test, which starts 144 hours after the end of the training data.

Table 5: Forecast accuracy over the 90-day test period, all areas (12,960 hourly observations).

Metric	**Baseline**	**Ridge**	**LightGBM**
MAE (MW)	0.525	0.305	**0.288**
RMSE (MW)	0.867	0.489	**0.426**
WAPE (%)	9.48	5.51	**5.20**
NMBE (%)	+0.87	−2.58	**+0.27**
MAE reduction vs. baseline (%)	–	41.9	**45.2**

Accuracy by price area. Table 6 shows that the advantage of LightGBM is concentrated in a few areas. It is clearly better in Oslo (0.64 MW against 0.81 MW) and better in Helsingfors, while Ridge has the lower MAE in Bergen, Stavanger and Tromsø, and the two are equal in Trondheim. Oslo is by far the largest portfolio (mean consumption 6.88 MW, Table 2), so its absolute errors dominate the pooled MAE and explain most of the overall difference. Measured by the scale-free WAPE, both models stay between 5.0% and 6.9% in all five Norwegian areas, against 9.4–12.1% for the baseline.

Table 6: MAE and WAPE by price area on the test period; the lowest value in each row is in bold.

	MAE (MW)			**WAPE (%)**			**MAE reduction (%)**	
Area	Base	Ridge	LGBM	Base	Ridge	LGBM	Ridge	LGBM
NO1 / Oslo	1.43	0.81	**0.64**	11.64	6.59	**5.16**	43.4	55.6
NO2 / Stavanger	0.61	**0.31**	0.37	10.59	**5.48**	6.49	48.3	38.8
NO3 / Trondheim	0.44	**0.24**	**0.24**	12.10	6.66	**6.52**	45.0	46.1
NO4 / Tromsø	0.18	**0.10**	0.11	9.40	**4.98**	5.89	47.0	37.3
NO5 / Bergen	0.36	**0.19**	0.23	11.24	**5.90**	6.94	47.5	38.2
FI / Helsingfors	**0.13**	0.18	0.15	**2.01**	2.77	2.27	−37.9	−13.3
All areas	0.53	0.31	**0.29**	9.48	5.51	**5.20**	41.9	45.2

Helsingfors. With a single fit, neither model beats the baseline in Helsingfors: the MAE increases by 38% for Ridge and by 13% for LightGBM, so the criterion of outperforming the baseline in each area where a model is applied (Section 1.4) would not be met there. As discussed in Sections 2.1 and 2.3, Finnish demand is stable and almost insensitive to temperature, so last week's value is already a good forecast, and only 3,617 training hours are available against 6,001 in the other areas. The retraining experiment below shows that part of this is a matter of data volume.

Effect of periodic retraining. To assess the effect of updating the models, we repeat the evaluation with refits every 7 days (weekly) and every day, each on all observations at least 144 hours old at that moment. This experiment is a separate training run and uses a LightGBM configuration from an earlier tuning run (learning rate 0.057, 69 leaves, 414 trees), so its frozen LightGBM results differ from Table 5 (MAE 0.300 MW against 0.288 MW); the percentages in Table 7 are relative to the frozen row of the same experiment.

Table 7: Single fit (frozen) versus periodic retraining on the test period, all areas.

Model	Training	MAE (MW)	RMSE (MW)	WAPE (%)	Bias (MW)	ΔMAE (%)
LightGBM	Frozen	0.300	0.432	5.42	+0.067	–
	Weekly refit	**0.271**	**0.410**	**4.89**	+0.073	−9.7
	Daily refit	0.273	0.413	4.93	+0.067	−9.0
Ridge	Frozen	0.305	0.489	5.51	−0.143	–
	Weekly refit	0.273	0.450	4.93	−0.045	−10.5
	Daily refit	0.276	0.462	4.98	−0.026	−9.5

Retraining improves both models by about 10%, and refitting daily brings no further gain over refitting weekly. The Ridge underforecasting bias nearly disappears once February and March enter the training data, and the two models end up within 1% of each other (0.271 MW against 0.273 MW). Table 8 shows that Helsingfors benefits most: with a weekly refit Ridge beats the baseline there (MAE 0.116 MW against 0.129 MW), while LightGBM only equals it (0.129 MW) and beats it with a daily refit (0.119 MW). These margins are small, and the choice of forecaster for Helsingfors is discussed in Section 4.6. We therefore recommend periodic retraining, at least weekly, in operation (Section 5.4).

Table 8: MAE (MW) by price area, single fit versus weekly refit; the lowest value in each row is in bold.

		LightGBM		Ridge	
Area	Baseline	Frozen	Weekly	Frozen	Weekly
NO1 / Oslo	1.432	0.626	**0.623**	0.810	0.690
NO2 / Stavanger	0.605	0.409	0.375	**0.313**	**0.313**
NO3 / Trondheim	0.440	0.221	**0.202**	0.242	0.208
NO4 / Tromsø	0.183	0.121	0.100	**0.097**	0.109
NO5 / Bergen	0.364	0.210	0.197	**0.191**	0.203
FI / Helsingfors	0.129	0.214	0.129	0.178	**0.116**

Error by hour of the day. Figure 5 shows that the baseline is least accurate around the start of the morning ramp (up to 0.61 MW at 05:00–06:00 UTC), while the models are much flatter (0.24 MW–0.34 MW). The frozen models have the highest errors, in particular Ridge from the morning to the evening (about 0.34 MW at midday), whereas the refitted models stay at or below about 0.30 MW. No hour is systematically badly forecast.

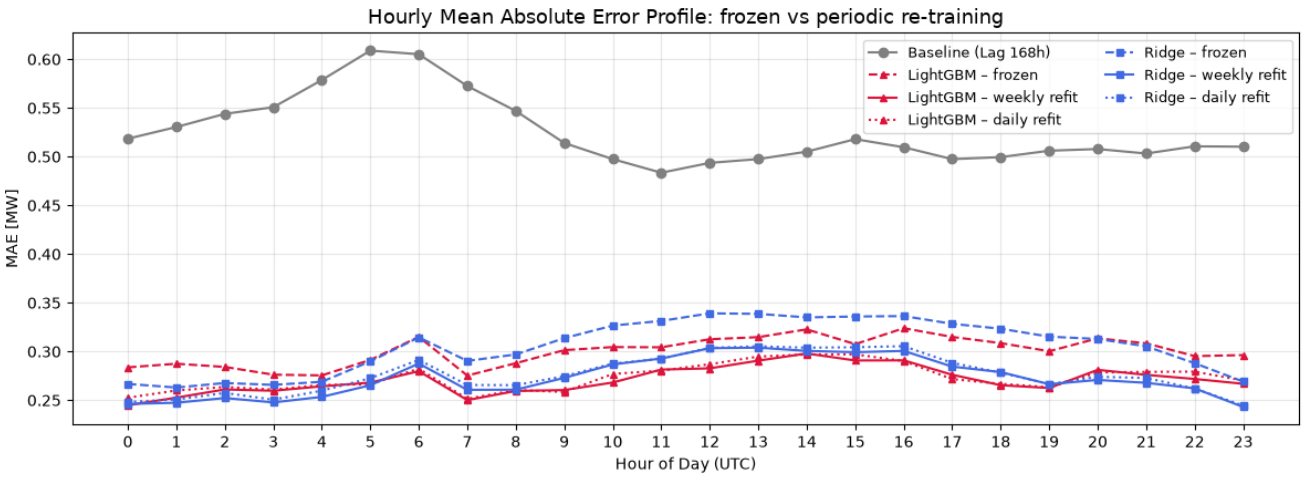


Figure 5: MAE by hour of the day (UTC) on the test period, all areas, for the baseline and for the frozen, weekly and daily refitted models of the retraining experiment.

Error in high-consumption hours. We define high-consumption hours as those at or above the 90th percentile of

test-period consumption in each area (1,301 observations). There the MAE is 0.768 MW for the baseline, 0.484 MW for Ridge and 0.384 MW for LightGBM, so LightGBM is 21% better than Ridge, against 5.6% over all hours. It has the lower MAE in Oslo (0.89 MW against 1.39 MW), Stavanger and Trondheim, while Ridge remains lower in Bergen, Tromsø and Helsingfors. All forecasters underforecast in these hours (mean error −0.71, −0.46 and −0.31 MW), partly because the hours are selected on high actual consumption.

Main drivers. In the LightGBM model the gain-based feature importance is dominated by consumption lags: the 144-hour lag accounts for 54% of the total gain, the 168-hour lag and the weekly rolling mean for 13% each, and the 192-hour lag for 9%. Each temperature and calendar feature contributes less than 1%, and the area variable 1.6%. Gain importance is only a rough guide for correlated features, but it shows that the model works mainly by adjusting recent consumption levels, with weather and calendar as refinements.

Assessment. Both models clearly improve on the baseline overall and in five of the six areas when fitted once. LightGBM is retained as the main model: in the main evaluation it has the lower MAE, RMSE and bias overall, and it is clearly better in the largest portfolio (Oslo). With periodic retraining the gap between the two models narrows to less than 1%. Ridge is a simpler, deterministic alternative that is better in three of the smaller areas, and it remains a credible choice.

3.7 Technical limitations

- **One year of data and a winter-only test.** The test period covers January to early April, so performance in summer or in other winters is unknown, and no multi-year seasonal validation is possible (Section 2.5).
- **The month feature.** January to March, which make up almost the whole test period, never occur in the training data, so a single fit must extrapolate to unseen months. Removing the feature worsens LightGBM but improves Ridge, whose month dummies for unseen months carry no information, so the single-fit Ridge results are probably somewhat pessimistic. Retraining mitigates this, except in April, observed in only 46 test hours per area (Ridge MAE from 0.57 to 0.67 MW with weekly refit).
- **Pooling and model class are confounded.** The global LightGBM and the local Ridge differ in both respects, so the comparison identifies which strategy performs better but not why, for instance in Helsingfors or Oslo. A local LightGBM and a global Ridge would be needed to separate the two effects; they were not estimated.
- **Seed sensitivity of LightGBM.** The LightGBM results come from a single run with a fixed seed, so that they can be repeated in the same software environment and that the technical and business evaluations share one set of forecasts (Section 4). Because of row and column subsampling, the result still depends on that seed. In a check with eleven seeds and the hyperparameters of Table 4, the single-fit test MAE varies between 0.284 MW and 0.308 MW (mean 0.295 MW, standard deviation 0.007 MW), and in Helsingfors it has a mean of 0.172 MW and a standard deviation of 0.032 MW. With a weekly refit the variation is small (0.270 MW ± 0.002 MW). LightGBM is therefore on average better than the deterministic Ridge model with a single fit (0.305 MW), but some seeds give a worse result, and the exact difference in the main table should not be over-interpreted. Averaging the predictions of several seeds would reduce this variability but was not done.
- **A single test window and no uncertainty estimate.** All results come from one 90-day split without confidence intervals or significance tests. The differences between the models in aggregate MAE and WAPE are small and within the seed variation, and only the larger differences (models against baseline, and the effect of retraining) should be considered reliable.
- **Tuning.** Forty trials is a small budget for nine hyperparameters, and early stopping on the validation folds makes the cross-validated MAE slightly optimistic. That MAE (0.551 MW) is much higher than the test MAE because the early folds are trained on only a few months of summer data, which makes the selection criterion favour models that extrapolate well. Both models are also trained on a squared-error loss, whereas the main metric is the MAE.
- **Single fit in the main evaluation.** The main results come from one fit that is not updated during the test. Periodic retraining improves accuracy by about 9–11% within the retraining experiment (Section 3.6), so the main tables understate what can be achieved in operation.
- **Temperature and uncertainty.** Temperature forecasts are taken as given and error-free, so real forecast errors would reduce accuracy. Only point forecasts are produced, without prediction intervals.
- **Helsingfors.** With a short history and low weather sensitivity, the margin over the baseline is small even when the models are retrained, and more data would be needed to confirm it.

4 Business evaluation, value and recommendations

Section 3 evaluated the models on forecast accuracy. This section asks what the differences mean for the retailer and for Aneo's service, using the Tier 2 and Tier 4 criteria of Section 1.4. All figures come from the same 12,960 test forecasts as Section 3.6 (90 days, 2 January to 2 April 2023, models fitted once), so the technical and financial results are directly comparable.

4.1 Basis of the evaluation

Three points define what the financial figures mean and how far they can be trusted.

- **Cost proxy.** Imbalance prices are not available (Section 1.5), so the cost of an error is its absolute size multiplied by the hourly spot price, as in Equation (7). The cost figures are therefore indicative, and the ranking of the models is more reliable than the absolute amounts.
- **Contract fee.** The fee is not given in the case. The ROI values of the evaluation imply a fee of NOK 900,000 for the test period (avoided cost divided by one plus the ROI ratio gives the same amount for both models). We treat this figure as an assumption and not as a fact about the contract.
- **Fixed models, winter only.** The models are fitted once and the test covers January to March only (Section 3.5). Section 3.6 shows that weekly retraining reduces the MAE by about 10%, so the figures below probably understate what the models achieve in operation. Tier 2 metrics for the retrained models were not computed, and we do not extrapolate them.

4.2 Financial value for the retailer

Table 9 compares the three forecasters. Both models are far better than the baseline. LightGBM is ahead of Ridge on most metrics, but the gap between the two models is much smaller than the gap to the baseline.

Table 9: Business metrics over the 90-day test period, all areas (Tier 2 and Tier 4). Rows marked † are derived from the reported values.

Metric	Baseline	Ridge	LightGBM
Total imbalance (MWh)	6,809	3,954	3,733
Avoided imbalance (MWh)	–	2,855	3,076
Reduction in imbalance volume (%)†	–	41.9	45.2
Imbalance cost (kNOK)	7,216	4,152	3,867
Avoided cost (kNOK)	–	3,064	3,348
Reduction in imbalance cost (%)†	–	42.5	46.4
P95 imbalance (MW)	1.89	1.10	0.89
P95/MAE(%)†	3.6	3.6	3.1
Trust, error $< 15\%$ (%)	76.2	96.5	95.4
Net service ROI (ratio; threshold ≥ 2)	–	2.40	2.72

LightGBM against Ridge: Both models reduce the total imbalance volume by more than 40%: LightGBM by 45% (3,076 MWh avoided) and Ridge by 42% (2,855 MWh). In money, LightGBM avoids NOK 3.35 million over the 90 days, about NOK 37,000 per day, and Ridge avoids NOK 3.06 million, about NOK 34,000 per day. The margin between the two models is approximately NOK 284k over 90 days: this is a non-trivial gap in absolute terms, though it is contingent on the quality of the spot-price proxy. Additionally, LightGBM has 221 MWh less imbalance and NOK 285,000 lower imbalance cost than Ridge. This is a real but modest advantage.

Risk and Trust: The P95 imbalance falls from 1.89 MW for the baseline to 1.10 MW for Ridge and 0.89 MW for LightGBM, which is 53% lower than the baseline and 19% lower than Ridge. The ratio of P95 to MAE is 3.1 for LightGBM and 3.6 for the other two forecasters, against about 2.5 for normally distributed errors (Section

1.4), so all forecasters have heavier tails than a normal distribution, but LightGBM has the lightest, meaning that its largest errors are less extreme compared to its average error. A lower P95 is what allows the retailer to carry a smaller risk premium in fixed-price contracts (Tier 4). The Trust metric shows the other side: the baseline is outside the 15% commercial threshold in about one hour in four (76.2% Trust), LightGBM in about one hour in 22 (95.4%) and Ridge in about one hour in 29 (96.5%). Trust is based on relative errors, so every area counts equally, whereas the MAE, the cost and the P95 are dominated by Oslo. This is why Ridge, which is better in the small areas, has the higher Trust.

Return on the service: With the assumed fee, the ROI ratio is 2.72 for LightGBM and 2.40 for Ridge, so both exceed the target ratio of 2 (Section 1.4). Since the target corresponds to a fee of at most one third of the avoided cost, the highest fee that still meets it is NOK 1.12 million for LightGBM and NOK 1.02 million for Ridge for a period of this length, and the retailer would break even at a fee equal to the avoided cost, NOK 3.35 million and NOK 3.06 million respectively.

4.3 Distribution across price areas

Table 10: Model comparison by price area over the 90-day test: accuracy (MAE, WAPE), risk (P95, Trust) and value (avoided cost, ROI). Bold marks the best of the three models on each metric. Avoided cost and ROI are relative to the baseline, so they are not defined for the baseline itself.

Area	Model	MAE (MW)	WAPE (%)	P95 (MW)	Trust (%)	Avoided cost (kNOK)	ROI (ratio)
NO1 / Oslo	Baseline	1.43	11.6	3.74	69.3	–	–
	Ridge	0.81	6.6	2.03	94.7	1,353	3.06
	LightGBM	**0.64**	**5.2**	**1.61**	**97.0**	**1,733**	**4.21**
NO2 / Stavanger	Baseline	0.61	10.6	1.44	71.5	–	–
	Ridge	**0.31**	**5.5**	0.90	**95.1**	**636**	**3.11**
	LightGBM	0.37	6.5	**0.87**	91.8	510	2.30
NO3 / Trondheim	Baseline	0.44	12.1	1.00	67.3	–	–
	Ridge	**0.24**	6.7	**0.54**	**97.1**	430	3.37
	LightGBM	**0.24**	**6.5**	0.56	95.4	**441**	**3.49**
NO4 / Tromsø	Baseline	0.18	9.4	0.45	79.9	–	–
	Ridge	**0.10**	**5.0**	**0.24**	**98.7**	**187**	**2.55**
	LightGBM	0.11	5.9	0.29	96.2	148	1.82
NO5 / Bergen	Baseline	0.36	11.2	0.82	70.7	–	–
	Ridge	**0.19**	**5.9**	**0.48**	**94.5**	**445**	**4.07**
	LightGBM	0.23	6.9	0.54	93.0	340	2.87
FI / Helsingfors	Baseline	**0.13**	**2.0**	0.54	98.5	–	–
	Ridge	0.18	2.8	**0.36**	98.9	−106	−1.61
	LightGBM	0.15	2.3	0.37	**99.2**	−37	−1.21
All areas	Baseline	0.525	9.5	1.89	76.2	–	–
	Ridge	0.305	5.5	1.10	**96.5**	3,064	2.40
	LightGBM	**0.288**	**5.2**	**0.89**	95.4	**3,348**	**2.72**

The area costs are computed separately and do not add up exactly to the portfolio total of Table 9 (the avoided cost of LightGBM is NOK 3.14 million summed over areas and NOK 3.35 million in the portfolio summary). We therefore use the portfolio figures for totals and the area figures only to compare areas and models. The ranking of the two models is the same on both bases.

Oslo. Oslo is the largest portfolio and therefore dominates both the aggregate MAE and the aggregate imbalance cost. LightGBM achieves an MAE of 0.64 MW against Ridge's 0.81 MW, a 21% advantage that translates directly into the largest share of the cost differential (+380kNOK) between the two models. LightGBM's non-linear tree splits are better suited to capturing Oslo's extreme winter peaks, where heating demand surges below 0°C and the consumption–temperature relationship becomes highly non-linear. In addition, Oslo carries 44% of the baseline cost and 55% of the avoided cost of LightGBM, and it is the only area where LightGBM is clearly better than

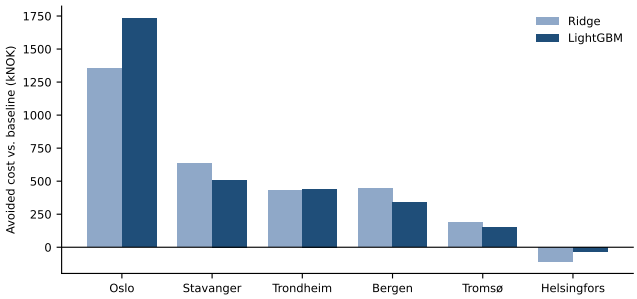


Figure 6: Avoided imbalance cost compared with the baseline by price area, 90-day test (kNOK).

Ridge: NOK 1.73 against 1.35 million avoided, a P95 of 1.61 MW against 2.03 MW, a Trust of 97.0% against 94.7%, and a ROI of 4.21 against 3.06.

Stavanger, Tromsø, Bergen. In terms of avoided costs, Ridge is better in Stavanger, Tromsø and Bergen, by NOK 269,000 together, and the two models are almost equal in **Trondheim.** Without Oslo, therefore, LightGBM would not be the better choice; its aggregate advantage is a result of one area. The lower WAPE values confirm that Ridge captures the seasonal pattern in these areas more efficiently.

Helsingfors. Neither model creates value in Helsingfors with a single fit: LightGBM costs NOK 37,000 more than the baseline and Ridge NOK 106,000 more. The baseline is already accurate there (WAPE 2.0%, Trust 98.5%): demand is almost insensitive to temperature because of municipal district heating, the series is almost flat (standard deviation 0.92 MW around a mean of 6.08 MW), so the seasonal naïve forecast already captures almost all of the variance (Section 2.3), and the models differ from it by less than one percentage point of Trust (LightGBM 99.2%, Ridge 98.9%).

These observations are weakened by the retraining experiment. With a weekly refit (Table 8), LightGBM achieves the lower MAE in Oslo, Trondheim, Bergen and Tromsø, and Ridge only in Stavanger and Helsingfors, and the overall MAEs are 0.271 MW and 0.273 MW. Most of the area-level advantage of Ridge on the frozen results therefore disappears once the models are retrained. For Helsingfors, the weekly Ridge refit has an MAE of 0.116 MW against 0.129 MW for the baseline and for LightGBM respectively, a gain of about 0.013 MW, or roughly 28 MWh over the 90 days.

4.4 Trade-offs

Accuracy and tail risk. LightGBM is better on MAE (0.288 against 0.305 MW), RMSE (0.426 against 0.489 MW), P95 and avoided cost, and slightly worse on Trust. Since the retailer's cost is driven by volume-weighted errors, and large errors are the most costly, we regard the volume-weighted metrics as the most relevant, as long as the Trust in the small areas is monitored (Section 6).

Cost of errors and bias. In our evaluation, an error costs the same whether the forecast is too high or too low. In practice the two directions may be priced differently, but we have no imbalance prices to check this. What we can check is whether a model makes mistakes mostly in one direction (bias). Ridge forecasts too low on average (NMBE −2.6%, about 1,855 MWh over the 90 days), so at least 47% of its imbalance goes in the same direction. LightGBM is almost unbiased (NMBE +0.3%, about 191 MWh, at least 5% of its imbalance). If underestimation is more expensive, Ridge could have a higher actual cost than reported. Weekly retraining reduces the Ridge bias from −0.143 to −0.045 MW (Table 7). If future settlement prices show that one type of error is more expensive, the models could be adjusted to favour the less costly direction.

Complexity and robustness. Ridge is simple and always gives the same result, but it needs six separate models, one per area. LightGBM is more complex inside, but it is one global model for all areas, so there is only one model to store, update and monitor. Its weak point is that a single fit depends on the random seed: the test MAE varies between 0.284 and 0.308 MW across seeds (Section 3.7). With weekly retraining this variation is only about 0.002 MW. We also considered using Ridge in Stavanger, Tromsø and Bergen and LightGBM elsewhere. On the frozen models this would add about NOK 269,000 over the 90 days, but we do not recommend it: the choice

would rest on one test period, where the differences in these areas are small (spanning from 0.01 and 0.06 MW, where 0.01MW is the difference of MAE between the two models for Tromsø and 0.06MW concerns Stavanger) and have no confidence intervals, and it would mean running three kinds of forecaster.

Interpretability. Ridge is easier to explain, because each coefficient shows directly how a feature changes the forecast. LightGBM relies mostly on past consumption: the lags and the weekly average account for about 89% of its importance (144 h: 54%, 168 h: 13%, weekly mean: 13%, 192 h: 9%), while temperature and calendar features contribute less than 1% each (Section 3.6). In simple terms, the model takes last week's consumption and adjusts it for the recent level. This is easy to explain, but it does not show the temperature effect that the retailer wants to see (Tier 4). We can add this with SHAP values, which show how much each feature pushed a forecast up or down, and by delivering the baseline and the temperature forecast together with each forecast.

Costs and viability. The costs of the service are development and maintenance effort, a weather-data subscription, computing (a small daily batch job, Section 5.2), monitoring and on-call time. No figures for these costs are available in the case, so the net value for Aneo cannot be computed. The retailer's side was derived in Section 4.2, where the fee must not exceed one third of the avoided cost. For Aneo, the fee F must also exceed its own cost C_{Aneo} over the same period, so the viable range is $C_{\text{Aneo}} < F \leq A/3$, where A is the avoided cost of Table 9. The table below shows how the ROI ratio moves with the fee.

Fee for 90 days (MNOK)	0.6	0.9	1.2	1.5
ROI ratio, LightGBM	4.58	2.72	1.79	1.23
ROI ratio, Ridge	4.11	2.40	1.55	1.04

The avoided cost is proportional to the assumed cost per MWh of error. At the assumed fee the ROI target of 2 requires an avoided cost of at least NOK 2.7 million (three times the fee). LightGBM achieves NOK 3.35 million and Ridge NOK 3.06 million (Section 4.2), so the target still holds if the true cost per MWh is down to 81% (LightGBM) or 88% (Ridge) of the spot-price proxy, and is missed below that. These shares are $3F/A$, that is the required avoided cost of NOK 2.7 million divided by the avoided cost of each model ($2.7/3.35 = 0.81$ for LightGBM and $2.7/3.06 = 0.88$ for Ridge).

Risks. The main risks to the value estimated above, and how the design addresses them, are summarised below.

Table 11: Main risks to the estimated value and how the design addresses them.

Risk	Effect on the value	Mitigation
Winter-only test	The savings of Table 9 may not hold in other seasons and should not be annualised.	Shadow mode that covers a summer period; tighter monitoring in the first summer (Section 5.5).
Cost proxy and bias	Over- and underforecasting may be priced differently. Ridge forecasts too low on average (NMBE −2.6%), so its real cost could be higher than shown.	Collect imbalance prices in shadow and advisory stages; adjust the models to err on the cheaper side (Section 4.4).
Temperature forecast errors	Temperature forecasts are assumed error-free (Section 2.5). Real errors can reduce accuracy in the cold hours, where costs are concentrated.	Archive forecasts with their issue time; compare against actuals in shadow mode (Section 5.5).
Data or pipeline failure	Falling back to the baseline gives up about NOK 37,000 of avoided cost per day in winter.	Fallback chain, quality gates and alerts (Section 5.3).
Fee below Aneo's cost	The service is not viable for Aneo even if the retailer benefits.	Agree the fee before the pilot; stop rule in Section 4.6.

4.5 Effect on the service KPIs

Retention and satisfaction cannot be computed from the dataset, so we only discuss how the results would affect them. The retailer is likely to value the service for three reasons, all shown in Table 9: a large saving compared with the fee, fewer large errors (a lower P95, which supports smaller risk premiums in fixed-price offers), and forecasts that are within the 15% threshold in most hours. These benefits last only if the forecast arrives every

day and can be explained. A day without the model forgoes, on average over the test period, about NOK 34,000 (Ridge) to NOK 37,000 (LightGBM) of avoided cost (Section 4.2), and forecasts that cannot be explained are harder to trust. Satisfaction therefore depends on both transparency and accuracy.

4.6 Recommendation

We recommend the following decision.

1. **Adopt the global LightGBM model for the five Norwegian areas (NO1–NO5), retrained at least weekly.** It has the lowest MAE, RMSE, P95 and bias overall, it gives the largest avoided cost (NOK 3.35 million, ROI ratio 2.72 at the assumed fee) and it is clearly better in Oslo, where the financial exposure is concentrated. With weekly retraining it is equal to or better than Ridge in four of the five Norwegian areas (Table 8) and within 1% overall, and a single global model is simpler to operate, version and monitor than six local ones.
2. **Serve Helsingfors with the seasonal-naive baseline.** With a single fit neither model beats the baseline there, and the weekly-refit gain of Ridge is about 0.013 MW with only 3,617 training hours. The baseline is transparent, needs no training, and removes a loss of NOK 37,000 compared with LightGBM on the frozen models (NOK 106,000 compared with Ridge). Adding this loss back to the portfolio figure gives approximately NOK 3.39 million avoided for the recommended policy (ratio about 2.76 at the same fee). The Tier 1 criterion of beating the baseline in every area (Section 1.4) is then met in the areas where a model is deployed, and Helsingfors is by definition equal to the baseline.
3. **Keep the baseline as the universal fallback, and run Ridge in shadow mode** (Section 5.5), because it is cheap, deterministic and better in Stavanger.

Next steps and decision rules. The recommendation is conditional on checks that the data available here do not allow: the Tier 2 metrics of the weekly-retrained models, the effect of real temperature forecast errors, real imbalance prices (including their asymmetry) and Aneo's own costs. We therefore propose to proceed through the stages of Section 5.5, each with a condition to continue and a condition to stop or change course.

Table 12: Next steps and decision rules by stage of the rollout.

Stage	Next step	Continue if	Stop or change if
Before the pilot	Agree fee, time zone and delivery format with the retailer; estimate Aneo's costs.	$C_{\text{Aneo}} < F \leq A/3$ (Section 4.4).	The fee is below Aneo's cost or above $A/3$: renegotiate or do not start (Section 4.4).
Shadow mode (several weeks; we propose at least eight)	Run LightGBM (NO1–NO5) and Ridge in parallel with the baseline. Collect imbalance prices, temperature-forecast issue times and the retailer's own forecasts.	The go/no-go criteria of Section 5.5 are met in an area.	In an area where they are not met, the area stays on the baseline (as Helsingfors). If this happens in all Norwegian areas, the models are not deployed.
Advisory delivery	Deliver the forecasts next to the baseline. Recompute the Tier 2 metrics for the weekly-retrained models with real imbalance prices.	The ROI ratio stays ≥ 2 at the agreed fee.	The ROI ratio falls below 2, or the avoided cost is negative in an area: that area returns to the baseline and the fee is renegotiated.
Production	Collect a second year of data. Review Helsingfors, the month feature and the choice between LightGBM and Ridge.	The gains persist and the monitoring metrics of Section 6 stay within their limits.	Performance degrades beyond the limits: retrain, or revert to the previous version or the baseline (Section 5.5).

5 Deployment and production readiness

This section describes how the recommended solution of Section 4.6 can be put into daily operation. The design follows from the constraints of the case: one batch forecast per day at about 10:00 for the 24 hours of the next day, consumption data delayed by about five days, and a temperature forecast for the target day as the only external input.

5.1 Architecture and required components

The service is a program that runs once a day, not a real-time system. Each run produces 144 forecasts (six areas times 24 hours), and the training set has 33,622 rows (Section 3.5), so no specialised infrastructure is needed. The service has six parts, listed in the order in which the data pass through them:

- **Ingestion and validation:** loads the consumption and the temperature forecast and checks them (Section 5.3).
- **Feature builder:** turns the raw data into the input variables of the model, with the 144-hour shift of Section 3.2.1. The same code is used for training and for the daily forecasts, so that the model receives the same kind of input in both cases. Differences between the two are a common source of errors in machine-learning systems (Sculley et al. 2015)
- **Forecast service:** computes the baseline for all areas first and the LightGBM forecast for NO1–NO5 afterwards, so that a fallback is always ready.
- **Model registry:** stores every model version together with how it was trained: data period, variables, settings, random seed and code version.
- **Quality gate and delivery:** chooses, for each area, LightGBM or the baseline (Section 5.3) and delivers the 144 values with time (UTC and local), area, forecast in MW, model used and model version. The retailer can therefore see which forecasts are fallbacks.
- **Archive and monitoring:** stores the inputs, forecasts and versions of every run and computes the errors when the actual consumption arrives. These errors are the input of the monitoring in Section 6.

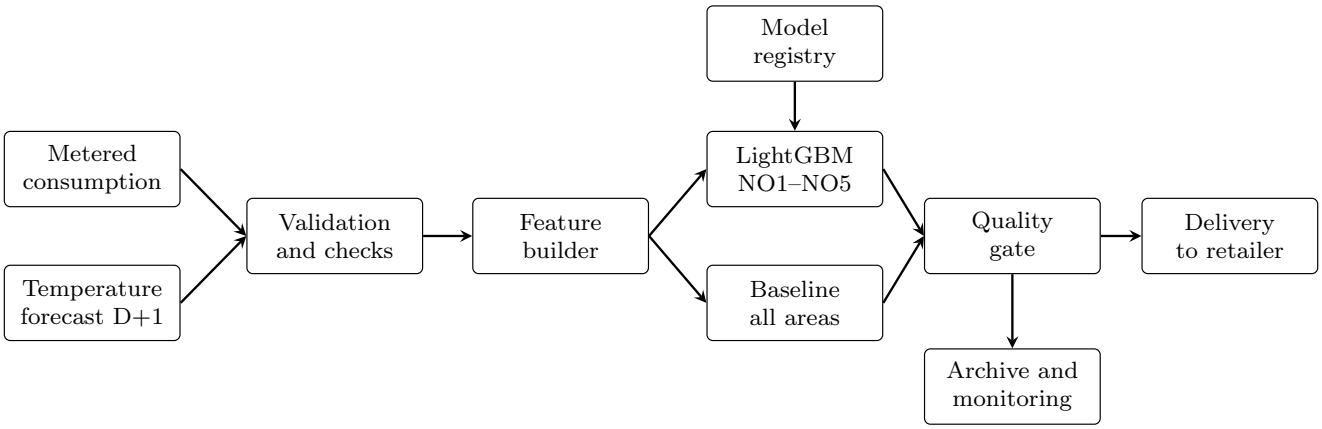


Figure 7: Daily forecasting pipeline.

Table 13 gives, for each component, a proposed implementation, the data it handles and who it depends on. The tools are proposals, since the case does not fix them, and must be agreed with Aneo's IT department and the retailer.

Table 13: Components of the daily forecasting service: proposed implementation, data and ownership.

Component	Proposed implementation	Interface and data	Dependency and owner
Ingestion and validation	Python script run daily by a scheduler	Consumption: hourly values per area from the retailer. Temperature: hourly forecast per location from the weather provider	Retailer, weather provider; Owner: Aneo data team
Feature builder and forecast service	Same Python code as in training	In: validated data. Out: 144 forecasts	Aneo data team.
Model registry	MLflow or a versioned folder	Model, training window, features, settings, seed, code version	Aneo data team
Quality gate and delivery	Same script; delivery by secure file transfer (SFTP) or REST API, whichever the retailer's trading system can read	Out: file with 144 rows (UTC and local time, area, forecast in MW, model, version, fallback reason)	Retailer's trading system; Aneo IT for the channel
Archive and monitoring	Database or file storage	Inputs, forecasts and versions of every run; errors added at day $D+6$	Aneo data team
Alerts	E-mail to the on-call person	Sent when a run fails or a delivery is missing	Aneo IT operations

The service depends on three external parties: the retailer for consumption data and for receiving the forecast, the weather provider for the temperature forecast, and a source for national holidays, loaded one year in advance (Section 5.2). Each dependency has a defined fallback in Section 5.3 (previous forecast run, baseline or last resort), so a late or missing input degrades the forecast but does not stop delivery. Delivery times, formats and responsibilities should be written into the agreements with the retailer and the weather provider before the shadow phase.

5.2 Daily schedule

Table 14 shows a proposed schedule for day D, in Central European time. The forecast must be delivered by 10:00, and the day-ahead auction closes at 12:00, so the pipeline is planned to finish at 09:30 and leaves a buffer to the deadline and a window for corrections. The schedule is a design proposal: neither the arrival time of the metered data nor the issue times of the temperature forecast are specified in the case, and they must be agreed with the retailer and the weather provider.

Table 14: Proposed daily schedule for day D (Central European time).

Time	Step
06:30	Load metered consumption up to the last hour of day $D-5$ for all six areas and validate it.
08:30	Fetch the latest temperature forecast for day $D+1$ (six locations, hourly) and store it as received, with its issue time (not available in the historical data, Section 2.4), so that it can be used in later retraining.
09:00	Build the features; compute the baseline for all areas and the LightGBM forecast for NO1–NO5.
09:15	Quality gate: choose model or fallback per area; log the reason for every fallback.
09:30	Deliver 144 hourly values to the retailer; archive inputs, outputs and versions.
09:45	If no delivery has been made, the stored baseline is sent, the main job is retried once and an alert is sent to the on-call person.
10:00	Delivery deadline (Tier 3). Automatic alert if no delivery has been confirmed.
10:00–11:00	Window for corrected re-issue; no changes after 11:00, so that the retailer can still use the forecast before the auction closes at 12:00.
$D+6$	When the metered data for $D+1$ arrive (five-day delay), compute the ex-post errors and update the monitoring metrics.

Two details must be considered. First, the delivery deadline is based on Norwegian time, while Finland is one hour ahead. The contract should clearly specify which time zone applies to Helsingfors.

Second, the dataset and the hourly analysis use UTC, while electricity demand follows local time. As a result, the same consumption peak appears one hour earlier or later in UTC when daylight saving time changes. The dataset includes the October 2022 change in the training set and the March 2023 change in the test set, but their impact on forecasting accuracy has not been evaluated. The pipeline should therefore be tested on both 23-hour and 25-hour days. Norwegian and Finnish public holidays, including holidays with changing dates, should also be loaded at least one year in advance. Finally, the forecasting job must run every day, including weekends and public holidays.

5.3 Data quality gates and fallback

The cleaning of the training data is described in Sections 2.4 and 3.2.1 and is not repeated here. This subsection covers what can go wrong in the daily run and what the service does about it. Since the dataset has no missing timestamps, the rules below cannot be derived from it; they are design proposals, to be validated by injecting faults. For a target hour on day $D+1$, the features need up to 14 days of history ending at 23:00 on day $D-5$ (Section 3.5), while the baseline needs only the same hour one week earlier, which is one day older.

Checks before delivery.

- **Consumption.** Every area must have a complete hourly history, without duplicates, zeros or flat stretches. Invalid values are treated as missing. Values outside the observed range of each area (Table 2) only raise a warning, because genuine extremes occur (Section 2.4). A gap of a few hours can be filled with the value of the same hour one week earlier, and the fill is logged.
- **Temperature forecast.** All 24 hourly values must be present for each area. If the latest run is missing or stale, the previous run is used. Because a feature uses the forecast of the previous day (T_{t-24}), that forecast must be archived too.
- **Forecast output.** Values must be finite, non-negative and below a ceiling based on the historical maximum of the area. A LightGBM forecast that differs from the baseline by more than the 99th percentile observed in the test period is flagged until it has been checked.

Fallback. If a check fails, the service does not stop: for each area and each day it switches to the next simpler forecast, because a simple forecast is better than none (Section 1.4). The steps are:

1. **LightGBM (main model):** used for the five Norwegian areas (NO1-NO5), provided that all input and output checks are passed.
2. **Previous LightGBM version:** if the new model has problems, the system uses the previous version, which has already been tested and approved.
3. **Baseline:** the consumption of the same hour one week earlier ($\hat{y}_{a,t} = y_{a,t-168}$). It is always used for Helsingfors, and for any other area when the latest day of consumption is missing, when no temperature forecast is usable, or when the forecast fails the output check. The seasonal-naive forecast is calculated in advance for all areas. This means that if LightGBM fails, the alternative forecast is already available and can be used immediately.
4. **Last resort:** if the data needed for the seasonal-naive method is also missing, the system uses the consumption from the same hour two weeks earlier. This is only an emergency solution, not a proper forecasting model.

Falling back has a price. In the test, the baseline had an MAE of 0.525 MW against 0.288 MW for LightGBM, which means giving up about NOK 37,000 of avoided cost per day on average, or roughly NOK 0.26 million for a week in winter (Section 4.2).

Failure of the pipeline itself. The steps above cover missing or invalid data, but not a run that does not start or crashes (for example, a scheduler or container failure). The baseline for day $D+1$ needs only consumption up to day $D-6$, which is already available at the run of day $D-1$. That run therefore stores the baseline for day $D+1$ in advance. If the main job has not delivered by 09:45, an independent delivery job sends this stored baseline, the main job is retried once, and an alert is sent to the on-call person. If the retry succeeds before the deadline, its forecast replaces the baseline. In this way the delivery of at least a baseline does not depend on the main job running correctly.

5.4 Retraining and model updates

- **Weekly refit.** The results show that retraining the model every week improves its performance. For LightGBM, the MAE decreases from 0.300 MW to 0.271 MW, an improvement of 9.7%. Daily retraining does not bring further improvements (MAE of 0.273 MW). Therefore, we recommend retraining the model once a week, using all the available data and keeping the same hyperparameters reported in Table 4. Hyperparameter tuning with Optuna could be performed every three months or when a drift alert is triggered.
- **Promotion gate for new versions.** A retrained model does not automatically become the new production model. Each week, a candidate is trained on all settled data except the most recent 14 days, which are used to compare it against the current model and the baseline. The candidate is approved only if (i) it passes the leakage test in Section 5.5, (ii) its MAE is lower than the baseline in every area, and (iii) its overall MAE is no more than 5% higher than that of the current model. If all conditions are met, the candidate is retrained using all available data, including the last 14 days, and becomes the new production model. Otherwise, the current model remains in use, the reason for rejection is recorded, and the model owner is notified. This ensures that the previous approved model is always available as a fallback, as described in Section 5.3.
- **Seed averaging.** LightGBM results can vary depending on the random seed used during training. In our experiments, the standard deviation of the MAE across different seeds was 0.007 MW. Training several models with different seeds and averaging their predictions could reduce this variation. This approach was not tested in our experiments and should be evaluated during the shadow-mode phase.
- **Event-based retraining.** In addition to weekly updates, the model should be retrained when important changes occur. These include changes in the input data, a drop in forecasting performance, or changes in the customer portfolio.
- **Review of the feature set.** The month feature may not generalise well because the dataset covers only one year. After collecting data from a second winter, we should check whether this feature is still useful or whether a smoother representation of yearly seasonality would work better.
- **Helsingfors.** For Helsingfors, the seasonal-naive baseline should remain in use until a new model proves to be more accurate. The weekly retrained Ridge model is a possible alternative, as it achieves an MAE of 0.116 MW compared with 0.129 MW for the baseline. However, it should first be tested in shadow mode over an agreed period.

5.5 Pre-deployment testing and staged rollout

Tests. According to the ML test score approach (Breck et al. 2017), the following tests are performed before the first delivery and every time the code or model is changed.

- *Leakage test:* checks automatically that no feature uses consumption data that is less than 144 hours old. This ensures that no future information is used (Tier 3, Leakage Pass) and is stricter than the 120-hour limit required by the case (Kaufman et al. 2012).
- *Replay test:* the system repeats the forecasts for each day of the test period, using only the data that was available at 09:00 on that day. The results must be close to those reported in Tables 5 and 6. Differences may indicate errors in the pipeline (Bergmeir and Benítez 2012).
- *Fault injection:* artificial problems are introduced to check that the fallback system works correctly and on time. These include missing or zero consumption data for day D − 5, missing or outdated temperature forecasts, and late data delivery.
- *Time and calendar tests:* check that the system works correctly during daylight-saving time changes, holidays, the start of a new year and the first forecast after a data gap.
- *Timeliness:* measures how long the entire pipeline takes to run and checks that it meets the required schedule.

Stages. The rollout is divided into four stages to reduce deployment risks and evaluate the models under real operating conditions before they are used for actual decisions.

1. **Offline validation** (completed): the models have been tested using historical data, including the weekly retraining experiment (Section 3). This provides an initial assessment of their performance, but does not fully reproduce real operating conditions.

2. **Shadow mode:** the new models are run every day in parallel with the existing baseline, but their forecasts are not used by the retailer. Instead, all forecasts are archived and compared with the actual consumption once it becomes available, approximately five days later. The baseline remains the model used for operational decisions throughout this stage. LightGBM is tested for the Norwegian areas, while Ridge is evaluated as a possible alternative, particularly for Helsingfors. This stage also allows us to evaluate seed averaging and check that the complete pipeline works correctly with real data and delivery conditions. It should continue for several weeks to collect enough observations.
3. **Advisory delivery:** after the shadow phase, the forecasts from the candidate models are delivered to the retailer alongside the baseline forecasts. They are provided as decision-support information, while the retailer continues to rely on its existing process to make operational decisions (Section 1.3). This allows the retailer to assess the usefulness and reliability of the new forecasts before they replace the baseline.
4. **Production:** once the models have demonstrated sufficient accuracy and reliability, they are activated for operational use. Helsingfors continues using the baseline, since the current experiments do not yet provide sufficient evidence to replace it.

Go/no-go criteria. Before leaving shadow mode, we check three things (Section 1.4): (i) in each area where the model is deployed, it beats the baseline; (ii) the leakage and timeliness checks pass; (iii) the error in the shadow phase is close to the error in the test (for example the WAPE of LightGBM in Table 5). For the third point we propose the following thresholds, to be agreed with the retailer before the shadow phase starts. The WAPE over the shadow period, overall and in each area, should not exceed the corresponding test value (Table 5) by more than 20% in relative terms (for example, at most 6.2% overall against 5.2% in the test). The Trust should not fall more than five percentage points below its test value (95.4%). The shadow phase should last at least eight weeks. These margins are a starting point: the week-to-week variation of the error in the test period (Section 3.6) should be used to check that normal fluctuations do not trigger a no-go, and that a real degradation is not accepted. If a threshold is not met in an area, the model does not go live there until the cause is found, and the area stays on the baseline. Because the offline test covers only the winter, the first months also do not validate the behaviour in summer, and the monitoring must be tightened during the first summer.

Going back after a problem. Every weekly retraining creates a new model version, which is stored in the model registry together with the earlier ones. If a new version causes problems, the pipeline returns to the previous approved version, that is, the one used in the week before. At the first go-live there is no previous version, so the service falls back to the baseline (Section 5.3). In both cases the switch only changes a configuration setting and needs no new deployment.

5.6 Reproducibility, versioning, security and scalability

Reproducibility and versioning. To make sure the results can be repeated, the environment is fixed with pinned versions of Python, scikit-learn, LightGBM and Optuna (Section 3.4). The random seed is fixed (42), and every registered model stores the training window, the feature list, the hyperparameters and the evaluation metrics. The original data and the final forecasts are also saved every time the model runs, so that any past forecast can be reproduced and checked. In particular, saving the temperature forecasts received is important because the original dataset does not record when these forecasts were made (Section 2.4). Using saved real forecasts for future training also makes the training process more similar to real operations. These practices follow common MLOps recommendations (Kreuzberger et al. 2023). The code is kept in version control and every registered model stores the identifier of the code version that produced it. A continuous integration job runs the leakage, replay and fault-injection tests of Section 5.5 on every change, and the container image used in production is versioned together with the code.

Security and privacy. The solution uses commercially sensitive data about the retailer's electricity consumption, and its forecasts affect the retailer's trading. Therefore, the data must be properly protected. Since the data are aggregated by price area and do not contain individual customer information, privacy risks are limited. However, whether personal-data regulations apply must be confirmed with Aneo's legal department. Access follows the principle of least privilege: only authorised members of the data science team and IT operations can access stored data and the model registry, while only the delivery job can send forecasts. Data are encrypted both when stored and when transferred, and passwords and other secrets are stored securely outside the code. Testing and production environments are also kept separate. Finally, access and forecast deliveries are logged to ensure that any changes

can be traced. Forecasts are sent through a secure channel and checked against the archived original files to detect possible changes or manipulation. The retailer and Aneo's legal department must also agree on where the data are stored, how they are processed and how long they are retained.

Scalability. The workload is small: 144 predictions per day and a training table of tens of thousands of rows. A single scheduled container is enough, and no distributed computing or GPU is needed. The training time and the daily runtime are not reported in the results, so they should be measured in the staging environment. The solution scales in the number of areas by adding the area as a category in the global model. The limiting factor is not computation, but the history needed for a new area (a week for the baseline, two weeks of history before the model can be scored, and enough data for a reliable fit), which is why new areas start on the baseline.

6 Monitoring and maintenance

The forecasting service must be monitored continuously to ensure that it remains accurate, reliable and valuable for the retailer. Even if the models perform well during testing, their performance may change over time due to changes in consumption patterns, weather conditions, the retailer's customer portfolio or the underlying data. Monitoring therefore covers data quality, model performance, business value and the operational reliability of the daily pipeline. The monitoring process builds on the architecture and fallback mechanisms described in Section 5.1.

6.1 Monitoring

The monitoring framework follows the four tiers of success criteria defined in Section 1.4. Since different indicators become available at different times, they are evaluated at different stages of the daily forecasting process.

Tier 1: Forecast quality Once the actual consumption data becomes available, approximately six days after the forecast, the prediction errors can be calculated. The primary monitoring metrics are MAE and WAPE, since they measure absolute and relative forecast accuracy. RMSE is used to identify unusually large errors, while NMBE reveals systematic over- or underforecasting. These metrics should be calculated both overall and separately for each price area, as aggregate performance can hide problems in smaller areas.

Two additional indicators are particularly relevant for the retailer. P95 imbalance measures the magnitude of errors in the worst 5% of hours, while Trust measures the percentage of hours in which the relative forecast error remains below 15%. Both should be monitored because a model can maintain a relatively stable MAE while becoming less reliable during individual high-risk hours. The current test results provide an initial reference: LightGBM achieved an overall MAE of 0.288 MW, a P95 of 0.89 MW and a Trust of 95.4%.

In addition to monitoring forecast errors, the input data must be checked for completeness, consistency and plausibility before each forecast run. The main checks include missing timestamps, duplicate records, unexpected zero values and unusually long periods of constant consumption. Temperature forecasts must be available for all six areas and all 24 target hours. Their issue time must also be checked to ensure that the latest available forecast is used. Furthermore, the distribution of important input features, particularly temperature and historical consumption, should be compared with the training data to identify potential changes.

Tier 2: Business value The technical performance of the models does not necessarily guarantee that the service continues to provide financial benefits. Therefore, avoided imbalance volume, estimated avoided imbalance cost and Net Service ROI should be recalculated once actual consumption is available. These indicators show whether improved technical performance translates into financial value for the retailer.

However, the estimated savings must be interpreted carefully. As discussed in Section 4, spot prices are only a substitute for imbalance prices and do not fully represent the retailer's actual settlement costs. Whenever actual imbalance prices become available, they should replace the current proxy. In addition, the assumed contract fee should be reviewed regularly to ensure that the calculated ROI still reflects the actual costs of the service.

Tier 3: Operational reliability Since the retailer needs the forecast before the day-ahead auction closes, delivery reliability is as important as prediction quality. The main operational KPIs are the percentage of forecasts delivered before 10:00, the number of failed runs, the frequency of fallback usage and the time required to complete the pipeline. The target remains 100% on-time delivery, including weekends and public holidays.

Every fallback must be recorded together with its reason and the model version that would otherwise have been used. This makes it possible to identify recurring problems, such as missing input data or repeated model failures. The system should also monitor whether the daily pipeline completes within the planned schedule, leaving sufficient time for corrections before the delivery deadline.

Tier 4: Customer relationship In addition to technical and financial indicators, the long-term success of the service depends on the retailer's satisfaction and willingness to continue using it. These aspects cannot be measured

directly using the available dataset. Instead, regular feedback from the retailer should be collected to evaluate whether the forecasts are useful for its trading decisions and whether the service meets its operational requirements.

Aneo should also maintain communication with the retailer to identify relevant changes in its customer portfolio, such as major customer acquisitions or losses. These changes can affect overall consumption patterns and may require adjustments to the forecasting solution. Customer feedback and contract renewal should therefore be added to the quantitative monitoring indicators.

6.2 Possible failures and warning indicators

Several factors can cause the forecasting solution to become less accurate over time. These can be divided into changes in the input data, changes in the relationship between input variables and consumption, and changes in the business environment.

Changes in the distribution of input variables may indicate that the conditions under which the model was trained are no longer representative. For example, unusually high or low temperatures, changes in seasonal patterns or shifts in historical consumption levels may affect forecast accuracy. This is particularly relevant because the dataset covers only one annual cycle. The model has therefore not been tested across multiple winters or different levels of seasonal variation. Monitoring should compare current input distributions with those observed during training and investigate substantial deviations.

A change in the input data does not necessarily mean that model performance has worsened. The more important indicator is whether forecast errors increase once actual consumption becomes available. A sustained increase in MAE or WAPE, a growing NMBE or a decline in P95 and Trust may indicate that the model no longer captures the underlying consumption patterns adequately. These metrics should be monitored over rolling periods to distinguish persistent degradation from normal short-term fluctuations.

One important limitation identified in Section 2.5 is that the available data does not contain information about the number or type of customers in the retailer's portfolio. If the retailer gains or loses customers, the overall consumption level may change without any corresponding change in weather or calendar conditions. The model could interpret such changes as forecasting errors rather than as a structural shift in the portfolio.

Aneo should therefore maintain communication with the retailer to identify relevant portfolio changes, such as major customer acquisitions or losses, and assess whether these require retraining or adjustments to the feature set.

Other external developments, such as changes in heating behaviour, energy prices or market regulations, may also affect consumption patterns. These changes cannot necessarily be identified from the available dataset alone. Regular discussions with the retailer are therefore an important addition to the automated monitoring.

6.3 Alerts and responsibilities

Monitoring indicators only become useful if deviations lead to appropriate actions. We therefore distinguish between critical alerts, which require immediate intervention, and performance warnings, which indicate a potential problem that should be investigated.

Critical alerts Critical alerts are triggered when the daily pipeline fails, required input data is missing or invalid, or the forecast has not been delivered by the 10:00 deadline. The existing fallback mechanism described in Section 5.3 should be activated automatically. The Aneo IT operations team receives an immediate alert and is responsible for investigating technical failures and ensuring that a valid forecast reaches the retailer.

Performance warnings Performance warnings are triggered when the forecast quality deteriorates over several evaluation periods. As an initial threshold, the WAPE should not exceed the corresponding test-period value by more than 20% in relative terms, either overall or in an individual price area. Similarly, Trust should not fall more than five percentage points below its test-period value of 95.4%. These thresholds follow the proposed go/no-go criteria in Section 5.5 and provide a consistent basis for monitoring after deployment.

A single unusual day should not automatically trigger retraining. Instead, rolling performance should be considered to avoid reacting to normal fluctuations.

In addition, the model must continue to outperform the seasonal baseline in every area where it is deployed. If this condition is no longer met over a sustained period, the model should be reviewed and the affected area should temporarily return to the baseline. A deterioration in P95 should also trigger an investigation, even if the average error remains acceptable, because unusually large errors can create substantial financial exposure.

Responsibilities Responsibility for monitoring is divided between the following stakeholders:

- **Aneo data science team:** Responsible for monitoring input distributions and forecast accuracy, evaluating model performance, investigating drift and deciding whether retraining or changes to the model are required. The team also maintains the model registry and documents all model updates.
- **Aneo IT operations:** Responsible for monitoring the daily pipeline, data ingestion, delivery confirmation and technical failures. The on-call person receives critical alerts and ensures that the fallback procedure is executed correctly.
- **Retailer:** Responsible for assessing whether the forecasts remain useful for its trading decisions and reporting relevant changes in its customer portfolio or operational conditions. The retailer should also provide feedback on the actual business value of the service.

This separation ensures that technical problems can be addressed immediately while changes affecting the business value of the solution are evaluated together with the end user.

6.4 Maintenance and model updates

The monitoring results determine which maintenance actions are necessary. The default maintenance strategy follows the weekly retraining schedule established in Section 5.4. Since weekly retraining improved LightGBM's MAE by approximately 9.7% compared with a frozen model, the production model should be updated once per week using all available settled consumption data. Daily retraining is not recommended, as it did not provide additional improvements in the experiments.

In addition to regular retraining, hyperparameter tuning should be performed approximately every three months or when a significant performance degradation is detected. Changes to the feature set should be evaluated periodically as well. In particular, the month feature should be reviewed after additional years of data have become available, since the current dataset only contains one observation of each month. A second winter would provide a better basis for assessing whether the observed seasonal patterns generalise.

Model updates and recalibration When a new model version is trained, it must pass the promotion criteria defined in Section 5.4 before replacing the existing production model. This includes outperforming the baseline in every deployed area and maintaining an overall MAE no more than 5% higher than the current model. If the candidate fails these conditions, the existing model remains in production.

Rollback and fallback If a deployed version causes unexpected problems, the service can immediately return to the previous approved version. If that version is also unavailable, the seasonal-naive baseline provides the fallback.

For Helsingfors, the baseline should remain in use until a more advanced model demonstrates a consistent improvement under real operating conditions. Although the weekly retrained Ridge model achieved a slightly lower MAE than the baseline in the experiment, the improvement is small and should first be confirmed through shadow-mode testing.

Continuous review Finally, the monitoring process itself should be reviewed regularly. Thresholds initially based on the 90-day test period may need adjustment once more operational data has been collected, particularly because the current evaluation covers only the winter months. The first summer after deployment should therefore

receive additional attention to determine whether forecast performance remains stable under different seasonal conditions.

Overall, monitoring and maintenance should ensure that the forecasting service does not simply retain its initial technical performance, but continues to deliver reliable forecasts and measurable value to the retailer. By combining automated quality checks, delayed performance evaluation, clear responsibilities and controlled model updates, Aneo can identify problems early while maintaining a reliable fallback and avoiding unnecessary changes to the production system.

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